A Lightweight Triple-Stream Network With Multisensor Fusion for Enhanced Few-Shot Learning Fault Diagnosis

Haotian Peng, Wei Wang, Jie Gao, Yu Wang, and Jinsong Du

***Abstract*—The application of multiple sensors significantly enhances the accuracy of industrial fault diagnosis, but existing algorithms are structurally complex and rely heavily on extensive training data. To optimize the efficiency of diagnosis, this article proposes a lightweight time-frequency-statistical domain fusion network. The model comprises three data streams that analyze the time-domain, frequency-domain, and statistical features of vibration signals, employing an improved channel attention mechanism for weighted fusion. In addition, two model-agnostic few-shot enhancement strategies are introduced, aiming to improve accuracy where training samples are scarce by reducing signal sample variations and optimizing the distribution of signals in the feature space. By combining these techniques, the proposed method exhibits superior performance in few-shot learning on two datasets compared to other multisensor fusion methods, while also achieving higher computational speed. The results of this research are of significant importance in enhancing the fault diagnostic capabilities of multisensor systems in practical industrial applications.**

***Index Terms*—Domain fusion, fault diagnosis, few-shot learning, multisensor fusion, rolling bearing.**

I. Introduction

PROGNOSTICS and Health Management is essential for the reliability of machinery and equipment [1], [2], especially the diagnosis of faults in critical components such as bearings that are susceptible to damage [3], [4]. By identifying minor equipment abnormalities as early as possible, and thus repairing and replacing them at an early stage of failure, financial losses, and human casualties can be prevented [5]. This approach utilizes accelerometers to capture vibration signals from equipment and data-based artificial intelligence methods to improve fault diagnosis accuracy. Huang et al. [6], [7] designed a deep adversarial capsule network and develop ensemble learning to intelligently decouple compound faults into single faults and generalize to new working conditions for industrial equipment. Li et al. [8] proposed a wavelet capsule network, an intelligent and interpretable machinery fault diagnosis framework, by leveraging the backward tracking technique. Chen et al. [9] developed a self-supervised fault diagnosis framework by integrating prior knowledge with deep learning to improve the accuracy and reliability of diagnosis models.

As a requirement to further improve diagnostic precision, more and more sensors are mounted at different locations on the equipment. In recent years, fault diagnosis methods that fuse signals from multiple sensors have become a new hotspot for research [10]. Fault diagnosis methods based on multisensor fusion strategies are mainly categorized into data, feature, and decision levels. Xiong et al. [11] used K-nearest neighbors (KNNs) and dimensionless indicators to assess the reliability of petrochemical machinery structures. They then apply evidence theory to fuse uncertain sensor data and make fault diagnosis decisions. Tong et al. [12] designed a multisensor information fusion method based on a kurtosis-weighted fusion strategy to enhance the shock characteristics of the sensor channel for interpretable bearing fault diagnosis. Tang et al. [13] proposed a minimum unscented Kalman filter-assisted deep belief network (DBN) for extracting invariant features from vibration signals acquired by multiple sensors by converting vibration signals into 2-D feature maps using Gramian angular summation fields. He et al. [14] used multisensor signals to pretrain and fine-tune a series of CNNs and finally obtained fault diagnosis results based on a decision fusion strategy with weighted majority voting. Yan et al. [15] proposed a CDTFAFN for machinery fault diagnosis. The signal-to-image encoding unit (SIEU), using ICQ-NSGT, converts raw vibro-acoustic signals into TFRs for coarse-grained feature fusion. However, these methods often require massive data as training samples to learn the fault feature associations among multiple sensors. In practice, labeled fault samples are often difficult to obtain in large quantities, which affects the practical application of these methods [16].

The objective of research on few-shot learning is to achieve good diagnostic accuracy through a small number of training

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Haotian Peng and Yu Wang are with the Shenyang Institute of Automation, Chinese Academy of Sciences, Shenyang 100169, China, also with the Liaoning Liaohe Laboratory, Shenyang 100169, China, also with the Key Laboratory on Intelligent Detection and Equipment Technology of Liaoning Province, Shenyang 100169, China, and also with the University of Chinese Academy of Sciences, Beijing 101408, China (e-mail: penghaotian@sia.cn; wangyu6@sia.cn).

Wei Wang, Jie Gao, and Jinsong Du are with the Shenyang Institute of Automation, Chinese Academy of Sciences, Shenyang 100169, China, also with the Liaoning Liaohe Laboratory, Shenyang 100169, China, and also with the Key Laboratory on Intelligent Detection and Equipment Technology of Liaoning Province, Shenyang 100169, China (e-mail: wangwei2@sia.cn; gaojie_sia@outlook.com; jsdu_sia@163.com).

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samples. Dong et al. [17] used a dynamic model to generate various bearing simulation data and implemented a few-shot learning fault diagnosis in real scenarios based on four parameter transfer strategies. Lei et al. [18] proposed an information fusion mechanism based on convolutional self-attention as a comprehensive decision-making module to solve cross-domain few-shot problems by meta-transfer learning embedded with prior knowledge. However, these methods often rely on transfer learning, which involves pretraining the network weights on another dataset. This introduces a risk: the effectiveness of fault diagnostic can be unstable when the dataset used for pretraining has significant feature differences from the small amount of learning data.

To explore the uniform fault feature representation under different vibration signal distributions, innovations in feature extraction are important. Yan et al. [19] fused frequency-sliced wavelet transforms to generate multisensor time–frequency representations and designed a fault diagnosis method based on adaptive multivariate feature pattern decomposition and multi-attention fused residual convolutional neural networks (CNNs). He et al. [20] extracted the low-dimensional features embedded in the high-dimensional nonlinear space of multisensor data and fused them into a 3-D pixel matrix, and proposed a novel intelligent diagnosis method based on two-scale residual networks. These methods commonly process the signal in the time–frequency domain using convolution, which starts with converting the 1-D vibration signal into a 2-D time–frequency map through different processing methods. Although these methods are verified to be effective in identifying fault features, the 2-D convolution brings exponential arithmetic consumption compared to simple 1-D operations, which is not conducive to enabling online fault diagnosis, especially in arithmetic-constrained terminals. This defeats the purpose of early detection of equipment faults.

To address the abovementioned challenges, this article proposes a lightweight triple-stream network with multisensor fusion for enhanced few-shot learning fault diagnosis. The main innovations of this approach can be summarized in two aspects.

1) *Design of a lightweight triple-stream network:* The proposed network allows feature extraction of signals from the time-domain, frequency-domain, and statistical features for integration toward fault diagnosis. It first performs multiscale analysis in the time and frequency domains separately and then fuses them using the attention mechanism, thus avoiding the complexity of feature extraction from 2-D time–frequency maps. The results are then combined with statistical metrics for output.
2) *Two model-independent few-shot learning enhancement schemes:* The difference in distribution between the training and test sets is reduced by optimizing the sampling interval. The intraclass and interclass distances in the feature space are used to calculate regular terms in the loss function to reduce the potential risk of overfitting. These methods enable high accuracy using few training samples without relying on any pretraining, avoiding the difficulties of transfer learning caused by the difference in prejtraining distributions.

The rest of this article is organized as follows. Section II reviews related works. Section III introduces the proposed lightweight triple-stream network and few-shot learning enhancement methods. Section IV validates the effectiveness through experiments on two publicly available datasets. Finally, Section V concludes this article.

II. Related Works

A. Multidomain Fault Features

Vibration signal is a function of amplitude with time as the variable, and feature extraction of time-domain signals is the most direct and widely used method. Liao et al. [21] used quadratic convolution to enhance the characterization of the model and effectively extracted fault features from the time-domain signals. Shi et al. [22] screened time-domain vibration signals with different distributions based on the boundary condition for sample mixing, to obtain smooth distributions and to enhance the generalization of the model.

For bearing faults, different locations and severities will excite vibrations with corresponding frequencies. These mechanism-based a priori information motivated related studies to feature extraction of vibration in the frequency domain. Wang et al. [23] used fast Fourier transform (FFT) to obtain the spectrum of vibration signals and input it into the graph structure. Lai et al. [24] simulated the frequency variation of bearing faults occurring and performed data enhancement on the spectrum obtained after FFT.

In addition, as the major fault diagnosis method before the rise of artificial intelligence, statistical features of vibration signals can effectively provide auxiliary diagnostic information with low computational requirements. Su et al. [25] extracted a variety of statistical features from vibration signal signals based on the a priori knowledge of the engineering system, which improves the accuracy of purely data-driven methods. Chen et al. [26] extracted the statistical features of a multisensor system and utilized a sparse autoencoder for effective feature fusion.

This article combines the vibration information from the three different domains mentioned above to extract fault features more comprehensively, which effectively improves the accuracy of fault diagnosis, especially when the number of training samples is small.

B. Multiscale and Channel Attention (CA)

Multiscale convolution is a widely used technique in CNNs to extract features at different scales by using convolution kernels of different sizes. Yu et al. [27] constructed a feature extractor using a multiscale residual network to enhance the accuracy of transfer learning. Chen et al. [28] built a multiscale split double-calibration network that combines period information to improve the accuracy of fault diagnosis in noisy conditions. Shao et al. [29] proposed an adaptive multiscale convolutional network to efficiently learn domain invariant features. Introducing multiscale convolution facilitates the identification of different frequency components in the vibration signals, which

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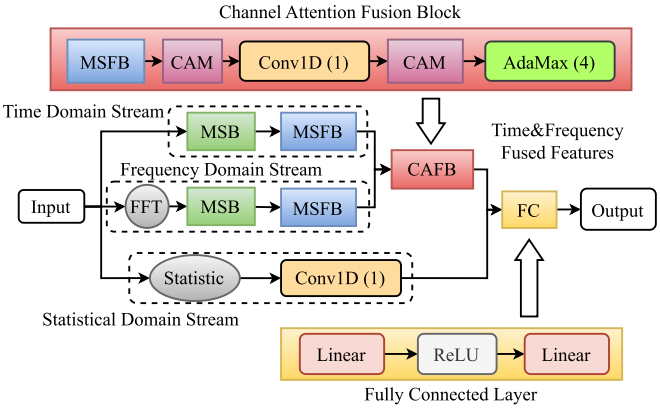


Fig. 1. Framework of the proposed triple-stream lightweight network.

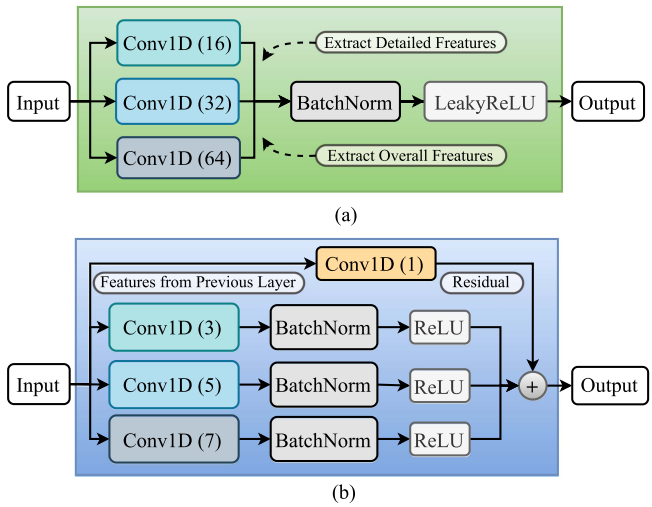


Fig. 2. Structure of modules in the proposed network. (a) MSB. (b) MSFB.

corresponds to different states and failure modes of mechanical equipment.

CA is an attention mechanism used in deep learning to enhance the contribution of different channels in the network to the overall performance. The goal of this mechanism is to allow the network to adaptively focus on specific channels, thereby improving the performance of the model. Hu et al. [30] originally introduced the concept of CA and proposed squeeze-and-excitation network (SENet) to implement it. Woo et al. [31] introduced the convolutional block attention module (CBAM), which sequentially integrates CA and spatial attention. The structure of the CA block is similar to that of the squeeze-and-excitation (SE) block, but it employs both max-pool and avg-pool, which greatly improves the expressive ability of the network. Wang et al. [32] proposed the efficient channel attention (ECA) block, which employs convolution to establish the interactions between the channels, replacing the MLP in the other CA.

Combining the advantages of CBAM and ECA, this article proposes a novel and ECA mechanism that dynamically assigns weights to vibration signal features of different scales and domains, enhances the feature representation capability, and achieves higher accuracy with less computation.

III. Methodology

The proposed method in this article consists of the following two aspects.

1) A lightweight triple-stream network, which extracts time, frequency, and statistical features from multisensor signals. Feature fusion is achieved through CA. This approach avoids 2-D operations on time–frequency maps, enhancing the network's efficiency.
2) Two network-agnostic few-shot enhanced approaches, involve reducing differences between samples and amplifying feature disparities from different classes.

A. Lightweight Triple-Stream Network

The architecture of the proposed lightweight triple-stream network is shown in Fig. 1. The network comprises three parallel streams, with each stream corresponding to the time, frequency, and statistical features of the signal. Initially, multiscale blocks (MSB) and multiscale fusion blocks (MSFB) are employed for multiscale feature extraction in the time and frequency domains separately, as well as for the fusion across multiple sensors. Subsequently, channel attention fusion blocks (CAFB) are utilized to merge time- and frequency-domain features. Finally, the concatenated statistical features and time–frequency fusion features are input into fully connected layers for output results.

1) Time–Frequency Decoupling Feature Extraction: Analyzing vibration signals in the time–frequency domain has proven to be beneficial for fault diagnosis, but this involves converting 1-D signals into 2-D signals, which leads to an exponential increase in computational overhead. To address this issue, this article uses multiscale convolutional layers to extract features from the time and frequency domains separately, thus reducing the computational burden.

After FFT, the time-domain and frequency-domain signals progress through MSBs, illustrated in Fig. 1. Within the MSB, three convolutional kernels of diverse sizes are employed to capture features across various scales within the signal (see Fig. 2). Subsequently, these features are concatenated, and normalized through BatchNorm, and the outcome is then propagated via the leaky rectified linear unit (LeakyReLU) activation function. The choice of LeakyReLU is motivated by the fact that the primary task of the MSB is to integrate information from multiple channels and output multiscale features. The results from these convolutional layers often exhibit a strong correlation with the original signal, and take on both positive and negative values. LeakyReLU prevents significant information loss that would occur with the use of the ReLU function.

The multiscale features extracted by the MSB are subsequently fed into MSFBs, as depicted in Fig. 2. The MSFB employs a residual structure, incorporating three convolutional layers with distinct sizes to further decompose the multiscale features of the signal. The outcomes of these layers are concatenated to form the residual component. Simultaneously, a convolutional layer with a kernel size of 1 is utilized to integrate additional multiscale features from the output of the

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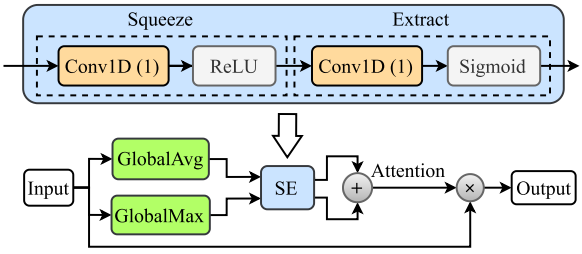


Fig. 3. Proposed CA structure. Both max-pooling and avg-pooling are used and interchannel links are established using convolution.

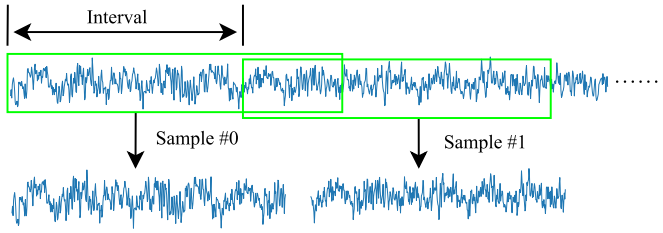


Fig. 4. Collection method of the vibration signal samples. The acceleration amplitude of a certain number of points is acquired at FIs.

preceding layer. Concurrently, this feature is reshaped to match the dimensions of the residual, serving as the shortcut connection. The integration of the residual structure plays a pivotal role in mitigating model degradation and enhancing overall network performance. The MSFB effectively facilitates the comprehensive integration of information from diverse channels, allowing for the extraction of multiscale features from the signal.

2) Attention-Based Feature Fusion: The time- and frequency-domain signals undergo multiscale feature extraction and interchannel information fusion through the MSB and MSFB, continuously. Subsequently, they are input into the CAFB for the fusion of time- and frequency-domain information. The key component of the CAFB is the channel attention module (CAM), as shown in Fig. 3.

Given an input feature map $X \in \mathbb{R}^{C\times L}$, where C and L represent the number of channels and the length of the feature map, respectively, the CAM is employed to extract the most important features from the input. The input feature map is first passed through a parallel combination of global average pooling (GAP) and global max pooling (GAM) by

$$z_{\text{avg}} = \frac{1}{L}\sum_{i=1}^{L}\mathbf{X}_i$$
$$z_{\max} = \max_{i=1}^{L}\mathbf{X}_i \tag{1}$$

where z_{avg} and $z_{\max}$ represent the results of GAP and GAM, respectively. Both of their sizes are $C \times 1$, representing global features for each channel

$$\begin{aligned} w_{\text{avg}} &= \sigma\left(\text{Conv}_2\left(\delta\left(\text{Conv}_1\left(z_{\text{avg}}\right)\right)\right)\right) \\ w_{\max} &= \sigma\left(\text{Conv}_2\left(\delta\left(\text{Conv}_1\left(z_{\max}\right)\right)\right)\right) \\ w &= w_{\text{avg}} + w_{\max} \\ X^* &= w \otimes X. \end{aligned} \tag{2}$$

As shown in (2), z_{avg} and $z_{\max}$ are individually input into a convolutional layer with shared weights. This layer has a kernel size of 1, which reduces the number of channels to capture interchannel relationships. The output size is $\frac{C}{r} \times 1$, where r represents the channel reduction ratio, a hyperparameter that can be specified. Following an ReLU layer, another convolutional layer with a kernel size of 1 is used to restore the output size to $C \times 1$. Finally, a sigmoid activation function is applied to map the output to the interval [0, 1], providing channel weights. The weights obtained from GMP and GAP are added up to form the overall weights. The weighted sum is then element-wise multiplied with the original feature map X, resulting in the feature map X^* after CA weighting.

Fig. 1 illustrates the composition of the CAFB. The time and frequency features are input into CAFB for further fusion, resulting in the output of time–frequency domain fused features. CAFB begins by employing an MSFB, facilitating the preliminary fusion of time–frequency domain features and concurrently extracting multiscale features. Subsequently, two CAMs are utilized to reinforce the weights of critical feature channels. A convolutional layer with a kernel size of 1 is inserted between the two CAMs for interchannel information fusion. Finally, AdaptiveMax is employed to output the most crucial features in each feature channel, contributing to the final classification results.

3) Statistical Feature Extraction and Fusion: In this article, we extracted 18 types of statistical features for the signal of each sensor, as detailed in Table I. Assuming the input multisensor signal is represented as $X \in \mathbf{R}^{C\times L}$, where C denotes the number of channels and L represents the signal length, and the statistical analysis operator is denoted as s, then $s(X) \in \mathbf{R}^{C\times 18}$. After the statistical analysis operator transforms the time-domain signal into statistical features, a convolutional kernel with a kernel size of 1 is used to fuse the statistical features across channels. The output is an overall statistical domain feature with a size of 1×18. The results of feature extraction in the statistical domain are concatenated with the fused time–frequency domain features and jointly output to a fully connected layer for fault diagnosis results. To facilitate the reproducibility of the paper's results, all the parameters of Conv1D are presented in Table II.

B. Enhanced Approaches for Few-Shot Learning

To further enhance diagnostic accuracy under few-shot settings, we propose two methods. First, an appropriate sampling interval is employed to maximize the similarity between samples, reducing the generation of ineffective sample features resulting from inappropriate sampling intervals. Second, a regularization loss term is designed, mandating the model to separate samples from different classes as much as possible in the feature space during the training process.

1) Minimize Intersample Variance: Fig. 4 illustrates the signal sampling process utilized in online diagnostics. In practice, a fixed-size time window is commonly employed for periodic sampling, with these samples feeding into the diagnostic model for real-time analysis. Rotating machinery such as bearings and

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TABLE I
STATISTICAL FEATURES USED IN THE PROPOSED NETWORK

Feature name	Expression	Feature name	Expression
Mean	$f_{\mathrm{m}} = \frac{1}{N}\sum_{i=1}^{N} x_i$	Peak-to-peak	$f_{\mathrm{p-p}} = f_{\max} - f_{\min}$
RMS	$f_{\mathrm{rms}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i)^2}$	Skewness	$f_{\mathrm{skew}} = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i - f_{\mathrm{m}})^3}{\left(\frac{1}{N}\sum_{i=1}^{N}(x_i - f_{\mathrm{m}})^2\right)^{\frac{3}{2}}}$
Variance	$f_{\mathrm{var}} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - f_{\mathrm{m}})^2}$	Shape factor	$f_{\mathrm{SF}} = \frac{f_{\mathrm{rms}}}{f_{\mathrm{ave}}}$
Square root amplitude value	$f_{\mathrm{sra}} = \left(\frac{1}{N}\sum_{i=1}^{N}\sqrt{\lvert x_i\rvert}\right)^2$	Crest factor	$f_{\mathrm{CF}} = \frac{f_{\mathrm{peak}}}{f_{\mathrm{rms}}}$
Absolute mean amplitude value	$f_{\mathrm{ave}} = \frac{1}{N}\sum_{i=1}^{N}\lvert x_i\rvert$	Impulse factor	$f_{\mathrm{IF}} = \frac{f_{\mathrm{peak}}}{f_{\mathrm{ave}}}$
Peak	$f_{\mathrm{peak}} = \max\lvert x_i\rvert$	Clearance factor	$f_{\mathrm{CLF}} = \frac{f_{\mathrm{peak}}}{f_{\mathrm{sra}}}$
Kurtosis	$f_{\mathrm{kurt}} = \frac{N\sum_{i=1}^{N}(x_i - f_{\mathrm{m}})^4}{(\sum_{i=1}^{N}(x_i - f_{\mathrm{m}})^2)^2}$	Frequency center	$f_{\mathrm{FC}} = \sum_{i=2}^{N}\dot{x}_i x_i \Big/ (2\pi\sum_{i=1}^{N} x_i^2)$
Maximum	$f_{\max} = \max(x_i)$	RMS variance frequency	$f_{\mathrm{RMSF}} = \sqrt{\sum_{i=2}^{N}\dot{x}_i{}^2 \Big/ (4\pi^2\sum_{i=1}^{N} x_i^2)}$
Minimum	$f_{\min} = \min(x_i)$	Root variance frequency	$f_{\mathrm{RVF}} = \sqrt{f_{\mathrm{RMSF}}^2 - f_{\mathrm{FC}}^2}$

TABLE II
PARAMETERS FOR CONV1D IN THE NETWORK

Block	Kernel	Stride	Padding	Out
MSB	16	16	0	8
	32	16	15	8
	64	16	31	8
MSFB[1]	1	2	0	24
	3	2	1	8
	5	2	2	8
	7	2	3	8
MSFB[2]	1	2	0	48
	3	2	1	16
	5	2	2	16
	7	2	3	16
* in Squeeze	1	1	0	4
* in Extract	1	1	0	48
* between CAs	1	1	0	48
* Fuse statistical features	1	1	0	1

[1] the MSFBs connected after MSBs.
[2] the MSFB inserted in CAFB.
* stands for the Conv1D

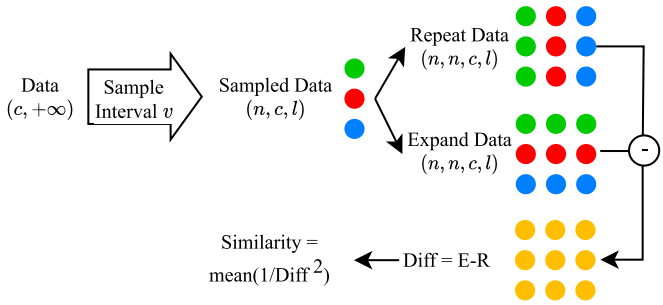


Fig. 5. Proposed method for calculating the similarity of samples at specific sampling intervals.

gears often generate highly periodic vibration signals. Building upon this prior knowledge, we opt to use the vibration signal's period as the sampling interval, aiming to reduce variations between samples introduced by the sampling process. This approach helps prevent the introduction of unsuitable signal features during the model's learning process, minimizes discrepancies between the training and test sets, and mitigates the risk of overfitting under few-shot settings.

The determination of the sampling interval involves computing the period length of the vibration signal. While conventional methods rely on calculating the spectrum and deriving the period from the main frequency, real-world vibration signals frequently exhibit significant noise that can impact frequency analysis. As a solution, this article introduces a direct method for period calculation, illustrated in Fig. 5.

Assuming the signal from the equipment is $X \in \mathbb{R}^{c\times\infty}$, where c is the number of signal channels. Let v be the sampling interval, and l be the sampling length. Sequentially sampling n samples on X, the sample set is denoted as $X \in \mathbb{R}^{n\times c\times l}$. Next, we repeat and extend the samples to obtain replicated data $R \in \mathbb{R}^{n\times n\times c\times l}$ and extended data $E \in \mathbb{R}^{n\times n\times c\times l}$. The repetition operation involves copying the entire sample set n times, while the extension operation duplicates each sample n times and concatenates them by

$$R[n,i,c,l] = x[i,c,l]$$
$$E[i,n,c,l] = x[i,c,l]. \tag{3}$$

Equation (4) calculates the sample similarity s_v at the sampling interval v. It subtracts E from R to obtain the differences between each pair of samples. The overall sample similarity is then computed from these differences, which is calculated in the same way as the mean squared error (**MSE**). Finally, the sampling interval v which maximizes the similarity s_v, is used as the optimal interval $\tilde{v}$ by

$$s_v = 1/\mathbf{MSE}(R,E)$$
$$\tilde{v} = \arg\max_{v} s_v. \tag{4}$$

To showcase our algorithm's effectiveness, we created a periodic waveform $y = 5\sin(\frac{2\pi}{7}x) + 3\sin(\frac{2\pi}{11}x) + 2\sin(\frac{2\pi}{17}x)$ mixed with noise $n\sim N(0,2)$. Fig. 6(a) demonstrates that noise poses challenges for FFT in extracting the underlying frequency information. In contrast, our algorithm in Fig. 6(b) distinctly reveals the 1309 $(7\times 11\times 17)$ periodicity and its double at 2618, effectively outlining individual subwave periodic characteristics. Fig. 6(c) and (d) shows the test results on a real dataset, where our proposed method yields the period of the vibration signal more efficiently than the FFT. Importantly, our algorithm involves only simple tensor operations and can be reduced to a qualitative analysis for int8 rather than a quantitative analysis

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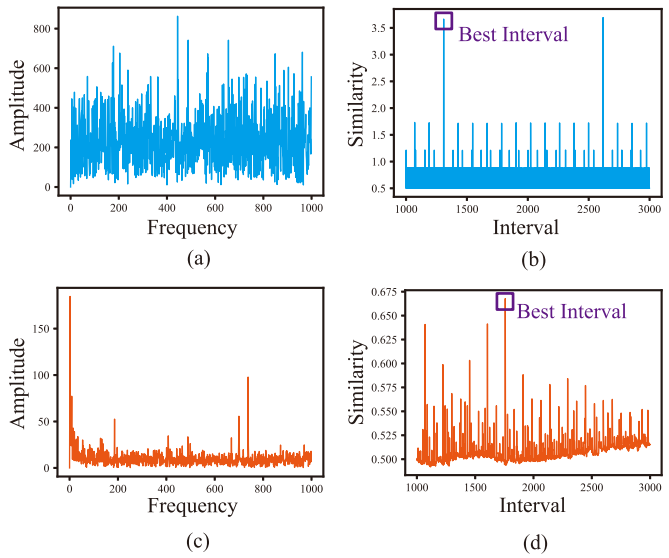


Fig. 6. Results of using different methods on simulated and real signals. (a) Simulated signal + FFT. (b) Simulated signal + Ours. (c) Real signal + FFT. (d) Real signal + Ours.

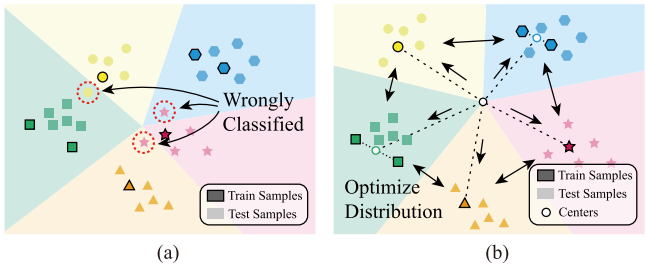


Fig. 7. Distribution of samples converging in feature space with different loss functions. (a) Cross entropy. (b) Adding the regularity term.

for float32, ensuring efficiency and practicality in real-world applications.

2) Maximize Interclass Variance: When using deep learning for fault diagnosis, the feature space is divided based on the training set features. During testing, the model localizes test sample features within this space to diagnose faults. The most commonly used cross-entropy loss function measures the difference between the training set output and the true label. This approach performs well with sufficient training samples but struggles under few-shot settings.

Fig. 7(a) illustrates the model's training, effectively mapping training samples to the feature space until the training loss hits 0, halting further training. However, during testing, some test samples land in incorrect spaces, reducing test accuracy compared to training and leading to overfitting.

To address this issue, we propose a loss regularization term (RT) designed to enhance the mapping of the feature space for a limited number of samples. By analyzing the aforementioned problem, the main issue is attributable to the limited number of training samples, rendering it impossible to fully represent the sample distribution of the test set. Moreover, since the sample distribution of the test set is unknown, the objective is to disperse and evenly distribute the training samples of different classes in the feature space, thereby increasing the probability of accurately mapping test set samples to the appropriate feature space, as shown in Fig. 7(b).

Therefore, it is necessary to output the position of the samples in the sample space before the sample feature output classifier, typically before the fully connected layer. Let the output result be $X \in \mathbb{R}^{b\times d}$, where b is the batch size, and d is the dimension of the feature space. Since the labels of the training set are known, we can use (5) to calculate the center position x_i of the ith class samples in the feature space, along with their average distance α_i from the class center

$$x_i = \frac{1}{n_i}\sum_{j=1}^{n_i} X_{ij}$$

$$\alpha_i = \frac{1}{n_i}\sum_{j=1}^{n_i} \|X_{ij} - x_i\| . \tag{5}$$

Assuming that this batch comprises samples from m distinct classes, we can utilize (6) to compute the central position c of these class centers as the centroid of the feature space. In addition, we can determine the distance β_i from the ith class center to the centroid of the feature space

$$c = \frac{1}{m}\sum_{i=1}^{m} \alpha_i$$

$$\beta_i = \|x_i - c\| . \tag{6}$$

Finally, utilizing (7), we calculate the intraclass average distance α, the mean distance β, and the variance γ from each class center to the centroid of the feature space

$$\alpha = \frac{1}{m}\sum_{i=1}^{m} \alpha_i$$

$$\beta = \frac{1}{m}\sum_{i=1}^{m} \beta_i$$

$$\gamma = \frac{1}{m}\sum_{i=1}^{m} \|\beta_i - \beta\| . \tag{7}$$

To minimize intraclass differences and maximize interclass differences, we calculate the RT, which is then added to the classification loss by

$$\mathcal{L}_{\text{reg}} = \alpha + \frac{1}{\beta} + \gamma$$

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda\mathcal{L}_{\text{reg}} \tag{8}$$

where $\mathcal{L}_{\text{cls}}$ is the categorization loss by cross-entropy and λ is a constant used to indicate the degree of regularization. This operation enables the model to achieve superior performance under few-shot settings, reducing the occurrence of overfitting.

IV. Experiments

A. Data Description

1) Dataset A: The Dynamic and Identification Research Group (DIRG) at Politecnico di Torino, Italy, has developed

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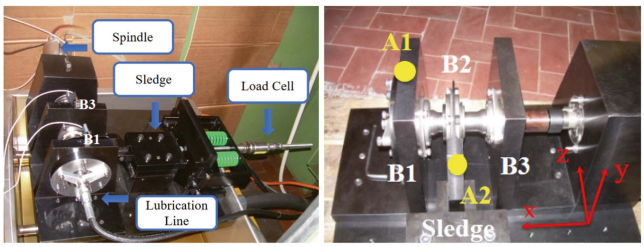


Fig. 8. Test rig and sensor install location for dataset A.

TABLE III
DETAILS OF WORKING CONDITIONS OF THE TWO DATASETS

Dataset	Condition	RPM	Load / N
A	C0	6000	0
	C1	6000	1000
	C2	12 000	0
	C3	12 000	1000
	C4	18 000	0
	C5	18 000	1000
B	C0	500	0
	C1	500	1500
	C2	1000	0
	C3	1000	1500
	C4	1500	0
	C5	1500	1500

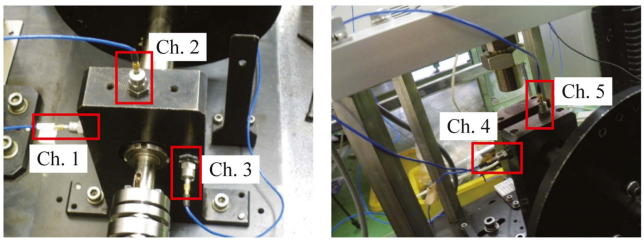


Fig. 9. Test rig and sensor install location for dataset B.

a rolling bearing test rig and openly shared the collected dataset [33]. The structure of the test bearing rig is depicted in Fig. 8. The test setup employs two triaxial Integrated Electronics Piezo-Electric accelerometer sensors. These two sensors are installed at two prominent positions on the bearing support structure, namely A1 and A2. Both accelerometers have a sampling frequency of 51.2 kHz and each sensor has 3 channels for a total of 6 channels.

In this dataset, faults in the rolling bearing are simulated by creating conical impressions on the inner ring or a single roller using a Rockwell tool. These 2 types of faults each have 3 different severities. In addition to normal conditions, all bearing signal states are divided into 7 $(1 + 2 \times 3)$ categories. By changing the load and speed of the test platform, we conducted experiments on data collected under 6 different working conditions (see Table III).

2) Dataset B: To further assess the learning capabilities of different algorithms in environments with extremely limited data, experiments are conducted on the dataset provided by Wang et al. [34]. The testing platform is illustrated in Fig. 9. Five accelerometers with different orientations are installed on the test rig, and the sampling frequency is 100 kHz. Due to its significantly higher sampling rate compared to other datasets, the duration of the same input points is notably short, posing challenges for the training of diagnostic algorithms.

In this dataset, artificial damage is performed on the outer ring, inner ring, and rollers of the bearing to simulate faults. Together with health status, all collected signals are divided into 4 categories. As with the DIRG dataset, we conducted experiments under 6 different working conditions by varying the load and rotational speed (see Table III).

B. Experimental Setup

1) Compared Methods: To assess the performance of the proposed time-frequency-statistical domain fusion network (TFS-DFN), we opt to compare it against seven state-of-the-art multisensor data fusion algorithms. These algorithms encompass diverse fusion strategies at the data, feature, and decision levels. The analyzed modalities include time, time-frequency, and statistical feature domains, facilitating a comprehensive evaluation of the efficacy of the proposed triple-stream network.

To elaborate on the comparative analysis, it is essential to underscore the significance of each fusion algorithm's approach. At the data level, fusion strategies involve combining raw sensor data, providing a foundation for subsequent feature extraction. At the feature level, algorithms integrate extracted features from individual sensors, aiming to capture diverse information for robust diagnostic decisions. Lastly, at the decision level, fusion methods consolidate the outcomes of individual sensors to formulate a final diagnosis.

1) *CNN:* We use CNN as a benchmark for comparison. It takes multichannel signals as inputs to the network, uses 1-D convolutional layers to extract features from the signals, and enhances the nonlinearity using max-pooling and ReLU.
2) *Wide deep convolutional neural network (WDCNN):* Zhang et al. [35] proposed a WDCNN characterized by the use of wider convolutional kernels in the first layer and smaller convolutional kernels in subsequent layers. Batch normalization is applied to all convolutional and fully connected layers to speed up the training process and improve the model performance.
3) *Improved convolutional neural network (ICNN):* He et al. [14] introduced an ICNN. They incorporate stochastic pooling into CNN, which samples all features based on their contribution probabilities and utilize Leaky ReLU as the activation function, helping address the vanishing gradient problem.
4) *Multisensor Adaptive Weighting Strategy (MSAWS):* Jiang et al. [36] proposed a method called MSAWS. This approach transforms time-domain signals into grayscale time–frequency maps through short-time fourier transform. Subsequently, an SE module is introduced to determine the importance of each single channel.
5) *Deep belief network based on Sparse Autoencoder (SAE-DBN):* Chen et al. [26] introduced an SAEDBN. This method extracts statistical features, inputting them into multiple two-layer SAEs for feature fusion. The fused feature vector trains a DBN for bearing fault classification.

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TABLE IV
COMPARISON OF DIFFERENT MODELS

Model	Fusion level	Signal domain	FLOPs	Params	Train	Eval
CNN	Data	time	2 157 264	144 391	0.152 ms	0.145 ms
WDCNN	Data	time	1 230 288	65 527	0.274 ms	0.247 ms
ICNN	Data	time	2 157 264	144 391	1.230 ms	1.230 ms
MSAWS	Data	time–frequency	115 910 944	5 230 347	2.429 ms	2.203 ms
SAEDBN	Feature	statistical	4494	4609	1.188 ms	1.151 ms
CFCNN	Feature	time	11 471 504	132 377	10.104 ms	10.028 ms
DSCNN	Decision	time	9 134 304	835 626	1.209 ms	1.081 ms
TFSDFN	Feature	time, frequency, and statistical	2 353 020	45 614	0.517 ms	0.433 ms

6) *Collaborative fusion convolutional neural network (CFCNN):* Xu et al. [37] proposed a CFCNN, which uses a multiscale shrinkage denoising module (MSDM) to extract multilevel features. Then, a central fusion module (CFM) is used to learn shared representations from multiple sources. (Note: The original method uses 4 stacks of MSDM and CFM modules, but the large number of parameters made it impractical under a few-shot setting, so only 1 set is used in the experiments).
7) *Dempster–shafer convolutional neural network (DSCNN):* Khan et al. [38] proposed a conflict resolution solution for multisensor decision-making based on the Dempster–Shafer (DS) evidence theory. A reward is applied to credible results, and a penalty is applied to noncredible results. To apply this to fault diagnosis, a DSCNN is established, using CNN to output results for each channel signal and utilizing this DS algorithm to output the final fusion result.

All models are compared in Table IV. Among all models, the proposed TFSDFN exhibits nearly the minimum parameter count and the fastest processing speed, providing significant advantages for training and online diagnostic capabilities. This lightweight and efficient design is typically conducive to enhancing training efficiency and real-time performance of the model, making it more competitive in resource-constrained environments.

2) Implementation Details: All of the mentioned methods are implemented using Python 3.10 and the PyTorch 2.0 deep learning framework on Ubuntu 20.04. The Adam optimizer is used for optimization with an initial learning rate of 1e−3. The learning rate scheduler employed is CosineAnnealingWarmRestarts, which combines cosine annealing and warm restarts. During training, the learning rate decays following a cosine function shape, while at the end of each cycle, the learning rate is reset through warm restarts to help escape local minima. Convergence is considered achieved when the minimum value reaches 5e−5. The first cycle consisted of 10 epochs, with each subsequent cycle doubling the duration of the previous one.

Regarding the dataset, 200 signal samples are collected for each fault under various working conditions. Each sample has a length of 2000, and the sampling interval is set to the period with the highest similarity among the signals. From these, 1, 5, and 10 samples are selected for each fault as the training set, and the remaining samples are evenly split between the validation and test sets. In the hyperparameter impact section of the experimental results, we also compared validation set settings with different ratios for reference.

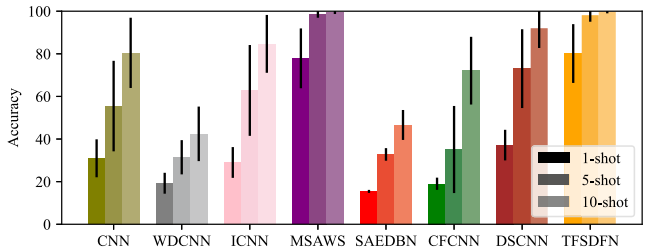


Fig. 10. Comparison of the accuracy of models on Dataset A with different few-shot settings.

Since the training set samples are randomly selected, to reduce the impact of the training samples selection on model training, each experiment is repeated 20 times. For each experiment, the same seed is used for random operations such as training data selection and model weight initialization, i.e., 1000, 2000,..., 20 000. Finally, the diagnostic performance of each model is recorded for each working condition and overall.

During the training process, each model is trained for 70 epochs using the training set data, and after each epoch, validation is performed using the validation set. After all training completion, the weights corresponding to the epoch with the highest validation accuracy are loaded. The model is then tested using the test set, and the results are used as the final evaluation metric for diagnostic performance.

C. Results and Analysis

1) Results on Dataset A: Table V illustrates the diagnostic accuracy of each model on Dataset A. The average diagnostic accuracy for each working condition is reported under 1-shot, 5-shot, and 10-shot training settings. The average accuracy of each model across different few-shot settings is also presented in the last column of Table V. Fig. 10 depicts the average performance and error range of different models under 1-shot, 5-shot, and 10-shot settings using a bar chart. By comparing the results, we can draw the following conclusions.

The lightweight time–frequency decoupling analysis can significantly reduce the number of parameters while maintaining accuracy: From the chart, it is evident that both the proposed TFSDFN and MSAWS exhibit superior performance. TFSDFN shows higher average accuracy and smaller error ranges under 1-shot and 10-shot settings, with comparable accuracy under 5-shot settings. However, TFSDFN has only 0.87% of the parameter count of MSAWS and requires only 2% of the computational resources.

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TABLE V
TEST RESULTS ON DATASET A UNDER DIFFERENT FEW-SHOT SETTINGS

Setting	Model	C0	C1	C2	C3	C4	C5	Avg.
1-Shot	CNN	23.86±1.8	23.84±2.1	26.59±2.2	39.76±3.4	45.50±3.8	26.21±2.5	30.96±8.9
	WDCNN	14.89±0.8	16.43±2.8	17.46±3.5	22.76±1.7	26.94±3.2	17.04±2.0	19.25±4.9
	ICNN	24.99±2.6	21.69±3.2	24.86±2.1	36.76±3.3	39.50±2.8	26.21±2.8	29.00±7.2
	MSAWS	71.61±3.8	56.56±6.2	89.63±2.4	79.16±6.1	99.05±0.8	73.39±3.0	77.88±14.0
	SAEDBN	15.29±0.6	15.36±0.7	15.31±0.9	15.77±0.6	15.20±0.4	15.63±0.7	15.43±0.7
	CFCNN	20.21±2.9	16.54±2.4	18.14±1.5	17.21±0.6	22.41±2.7	19.57±1.4	19.02±2.9
	DSCNN	33.06±1.5	29.46±3.5	35.51±3.7	44.53±2.3	48.21±3.4	33.26±1.7	37.15±7.2
	TFSDFN	**86.86±5.6**	**57.94±2.8**	**93.02±3.4**	**75.47±3.7**	**96.35±1.3**	**71.14±2.6**	**80.13±13.8** (+2.9%)
5-Shot	CNN	28.90±6.5	37.22±8.0	60.74±8.2	66.87±8.8	88.45±5.2	50.66±11.0	55.47±21.2
	WDCNN	21.25±2.0	28.73±1.3	30.29±1.0	34.96±0.8	46.18±4.0	27.27±2.1	31.45±8.0
	ICNN	38.29±5.9	37.29±7.5	67.58±3.3	82.65±5.1	92.57±3.3	58.48±3.7	62.81±21.3
	MSAWS	98.89±0.8	**95.71±1.3**	**99.78±0.3**	98.40±1.2	99.93±0.2	98.76±1.0	**98.58±1.6**
	SAEDBN	30.87±2.1	33.69±1.9	31.55±3.2	31.41±1.8	33.03±1.8	36.09±3.0	32.77±3.0
	CFCNN	21.79±2.9	19.66±6.6	40.92±20.1	39.56±19.3	66.47±6.3	22.11±4.0	35.09±20.4
	DSCNN	47.12±5.9	57.47±7.0	76.25±9.4	92.51±3.8	95.16±3.4	69.73±7.0	73.04±18.5
	TFSDFN	**99.80±0.2**	91.68±2.0	98.95±0.6	**99.45±0.3**	**100.00±0.0**	**99.11±0.5**	98.13±3.1 (-0.5%)
10-Shot	CNN	54.17±5.3	66.60±7.5	84.65±6.3	95.41±3.5	98.67±1.2	83.17±5.5	80.45±16.5
	WDCNN	26.63±1.3	36.30±3.0	40.56±1.4	47.88±1.9	66.86±2.3	36.39±2.6	42.44±12.8
	ICNN	62.38±6.0	73.14±4.1	89.67±1.5	97.20±1.3	99.67±0.3	85.95±3.2	84.67±13.6
	MSAWS	99.83±0.1	98.32±1.1	99.88±0.1	99.46±0.9	**100.00±0.0**	99.47±0.4	99.49±0.8
	SAEDBN	51.19±6.7	47.94±5.8	45.23±3.3	41.11±2.5	53.98±6.8	40.21±1.8	46.61±7.0
	CFCNN	55.49±5.3	62.41±6.5	86.32±4.8	80.84±5.9	91.97±1.5	55.49±8.1	72.09±15.9
	DSCNN	75.50±7.4	88.50±5.5	96.34±2.1	99.31±0.6	99.38±1.2	93.02±3.5	92.01±9.3
	TFSDFN	**100.00±0.0**	**98.33±0.9**	**99.94±0.1**	**100.00±0.0**	**100.00±0.0**	**100.00±0.0**	**99.71±0.7** (+0.2%)

Bold values indicate the best metric value under the same experimental setup, while underlined values indicate the second-best.

The proposed TFSDFN outperforms other models in all scenarios: Except for the significantly parameter-heavy MSAWS, the second-best performing model is DSCNN. Under the 1-shot setting, TFSDFN achieves an average diagnostic accuracy of 80.13%, surpassing DSCNN by 43%. Under the 5-shot setting, TFSDFN achieves an average diagnostic performance of 98.13%, outperforming DSCNN by 25%. Under the 10-shot setting, TFSDFN's accuracy is 8% higher than DSCNN.

Under the few-shot settings, the proposed TFSDFN exhibits more pronounced advantages: While the compared models perform well with 10 shots or more, there is a significant decrease in accuracy under 1-shot and 5-shot settings. This suggests that these algorithms focus on summarizing patterns of multiple sensor fusion from a large amount of data. In contrast, TFSDFN maintains stable accuracy under 1-shot and 5-shot settings, demonstrating its ability to effectively extract signal features for diagnosis under few-shot settings.

2) Results on Dataset B: To conduct a more comprehensive evaluation of the proposed TFSDFN and compare its accuracy with other multisensor fusion algorithms, we conducted similar experiments on Dataset B, which contains less information within the same data length, rendering the achievement of satisfactory diagnostic accuracy more challenging, particularly under few-shot settings. Table VI provides detailed results for various models under 1-shot, 5-shot, and 10-shot settings. In addition, Fig. 11 visually compares the performance of different models using bar charts. Analyzing the experimental results, we can draw the following conclusions:

The superior accuracy of TFSDFN in extreme environments: Dataset B contains less effective information, posing a more challenging evaluation for model performance. Experimental results demonstrate that TFSDFN consistently outperforms other comparative models under all testing conditions. In contrast

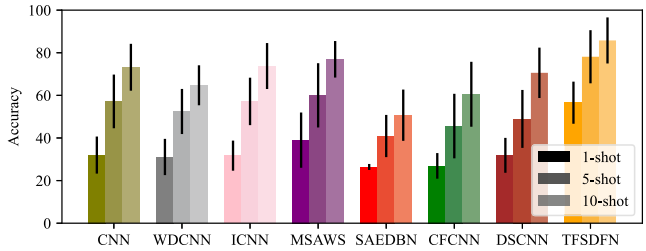


Fig. 11. Comparison of the accuracy of models on Dataset B with different few-shot settings.

to MSAWS, which has a complexity two orders of magnitude higher, TFSDFN achieves higher average accuracy across all conditions and maintains greater stability, especially under settings with very few samples such as 1-shot and 5-shot.

The superiority of TFSDFN in few-shot learning: In comparison to other models, TFSDFN demonstrates the ability to maintain consistently high accuracy even under few-shot settings. Under the 1-shot setting, it achieves performance comparable to some other models trained under the 10-shot setting. Even in the context of extreme environments, as represented by Dataset B, TFSDFN achieves a 78.12% accuracy under the 5-shot setting, highlighting its potential value in practical fault diagnosis applications.

TFSDFN demonstrates broad applicability: To provide a more intuitive comparison between TFSDFN and other models, we visually represent the accuracy of each model across various working conditions using radar charts, as shown in Fig. 12. Each subgraph displays the accuracy of six distinct working conditions. The optimal results of the comparative models for each condition are depicted in blue, while the average accuracy of all comparative models is represented in orange. TFSDFN

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TABLE VI
TEST RESULTS ON DATASET B UNDER DIFFERENT FEW-SHOT SETTINGS

Setting	Model	C0	C1	C2	C3	C4	C5	Avg.
1-Shot	CNN	26.01±3.4	27.60±3.8	31.36±7.3	28.52±5.2	32.88±6.4	45.49±7.4	31.98±8.7
	WDCNN	26.30±2.2	26.11±2.6	27.84±3.1	28.89±6.2	32.89±7.2	44.43±8.8	31.08±8.5
	ICNN	26.88±3.2	29.80±3.6	30.07±4.4	27.73±2.1	33.09±5.7	42.68±7.2	31.71±7.1
	MSAWS	26.84±4.2	27.87±3.2	41.63±10.6	34.31±9.5	51.25±8.2	52.00±9.6	38.98±13.0
	SAEDBN	26.50±1.3	26.02±1.0	26.45±1.7	26.07±1.2	25.95±1.6	27.40±0.7	26.40±1.4
	CFCNN	25.16±2.6	25.48±4.8	26.81±4.7	27.26±5.3	25.70±5.1	31.00±9.3	26.90±6.0
	DSCNN	25.59±2.1	26.79±2.6	30.61±5.3	31.21±4.4	33.76±6.2	43.17±10.3	31.86±8.2
	TFSDFN	**41.40±3.5**	**51.00±2.5**	**54.85±1.8**	**56.00±2.9**	**65.55±1.5**	**70.55±3.6**	**56.56±9.9** (+45.1%)
5-Shot	CNN	47.93±9.4	55.24±5.3	56.40±14.4	50.19±10.4	61.54±9.6	71.63±7.8	57.16±12.6
	WDCNN	39.50±5.6	44.62±4.1	53.59±3.6	47.10±4.1	62.66±1.9	67.11±3.5	52.43±10.6
	ICNN	42.92±7.5	51.35±3.4	56.97±10.6	57.16±5.2	61.75±4.0	72.80±4.3	57.16±11.1
	MSAWS	35.79±7.6	49.12±6.8	63.74±5.4	64.08±6.0	71.27±7.0	76.07±4.9	60.01±15.1
	SAEDBN	28.08±3.3	35.87±4.7	38.30±4.3	40.36±4.6	54.20±6.6	48.69±6.3	40.92±9.9
	CFCNN	34.57±5.3	38.14±11.6	55.58±14.3	45.34±10.8	45.74±17.4	54.15±15.4	45.59±15.1
	DSCNN	33.84±8.6	43.24±9.6	43.88±6.2	53.95±7.7	48.66±7.0	70.11±6.0	48.95±13.6
	TFSDFN	**58.52±4.1**	**67.27±2.4**	**85.13±2.2**	**75.61±1.8**	**91.79±1.4**	**90.43±2.2**	**78.12±12.5** (+30.2%)
10-Shot	CNN	59.16±3.8	64.47±7.1	80.41±7.4	70.66±5.5	80.16±7.5	84.28±4.2	73.19±11.0
	WDCNN	53.71±3.0	54.89±2.7	70.50±3.1	59.34±2.1	76.07±1.6	73.86±1.6	64.73±9.4
	ICNN	58.86±2.3	66.45±5.1	84.71±3.8	67.83±3.1	81.67±7.4	83.07±2.6	73.76±10.8
	MSAWS	63.93±7.3	73.60±8.7	80.57±3.0	77.30±3.7	82.28±2.4	83.88±3.7	76.93±8.6
	SAEDBN	33.01±3.7	42.92±3.6	50.50±7.2	50.09±4.9	64.79±5.3	62.82±4.4	50.69±12.0
	CFCNN	41.64±7.5	51.24±10.7	63.08±12.6	61.14±11.2	70.22±11.7	75.75±5.0	60.51±15.2
	DSCNN	52.36±8.0	64.30±4.5	73.11±5.9	75.25±4.0	75.03±7.9	83.45±7.5	70.58±11.8
	TFSDFN	**67.13±1.9**	**77.95±2.4**	**92.76±1.8**	**84.53±2.3**	**97.63±0.8**	**94.66±2.0**	**85.78±10.8** (+11.5%)

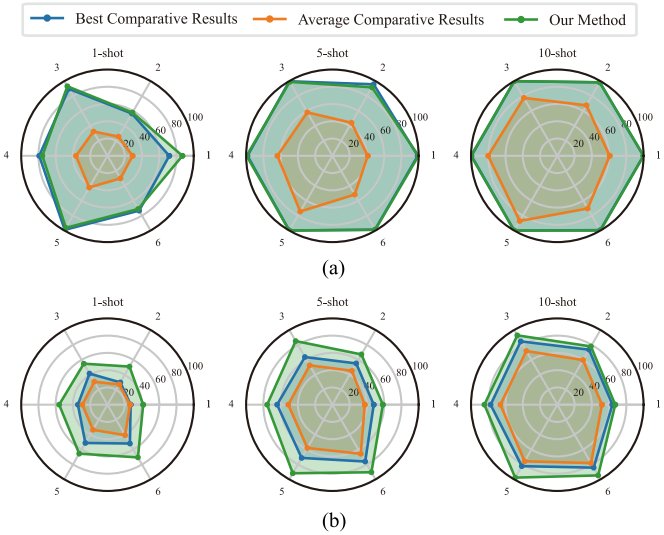


Fig. 12. Our proposed method has optimal few-shot learning capability under all working conditions. (a) Accuracy comparison across working conditions on Dataset A. (b) Accuracy comparison across working conditions on Dataset B

consistently exhibits the highest accuracy across all settings, whether in ordinary settings represented by Dataset A or in extreme settings represented by Dataset B. In addition, TFSDFN demonstrates a stable accuracy across different working conditions, avoiding inefficiencies in specific working conditions.

3) Impact of Hyperparameters: For deep learning models, the hyperparameter settings can significantly impact model performance. We conduct a sensitivity analysis on various hyperparameters to demonstrate the robustness of the model. The experimental results are presented in Fig. 13.

Batch size: The batch size determines the number of samples fed into the model during each training step. A batch size that is too small may lack samples from different categories within the same batch, potentially affecting the convergence direction of the loss function. This could lead to slower training and reduced accuracy. Since this study focuses on few-shot learning with a small-scale training set, we can feed the entire training set into the model at once to achieve optimal performance.

Learning rate: The learning rate determines the magnitude of weight updates during each training iteration. Experimental results show that both excessively large and small learning rates can severely affect the convergence of the model. We select a learning rate of 1×10^{-3} as the optimal value.

Degree of regularization: An RT is introduced to enhance the model's accuracy under few-shot learning conditions. Experimental results indicate that under extremely limited training samples (e.g., 1-shot), an overly strong RT can reduce accuracy. This is because the model prioritizes optimizing the feature distribution of the samples rather than focusing on the fault classification task. We select 1×10^{-5} as the optimal degree of regularization.

Validation set ratio: For the training set, we select 1, 5, and 10 samples per fault category to represent different few-shot learning scenarios. Experiments are conducted to investigate the impact of various validation set ratios on the test set accuracy. Due to the optimized sampling strategy, differences between samples are minimal, and varying the validation set ratio does not lead to significant performance differences. A ratio of 0.5 is selected as the final choice.

4) Robustness and Generalization: To validate the robustness of the proposed method, we introduce different levels of noise into the training data and evaluate the model's accuracy, as shown in Fig. 14. Following similar approaches in existing studies, additive Gaussian noise with a mean of 0 is used to simulate environmental noise, and the signal-to-noise ratio (SNR) is employed to quantify the noise intensity. For

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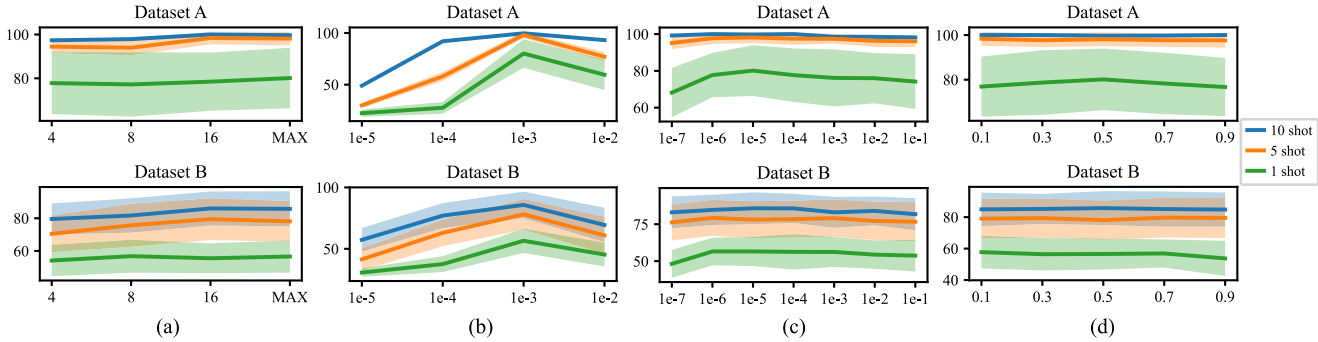


Fig. 13. Trends in the accuracy of few-shot learning under different hyperparameter settings. (a) Different batch sizes, where MAX represents treating the entire training set as a batch. (b) Different learning rates. (c) Different degrees λ of regularization. (d) Different validation set ratio of the whole dataset.

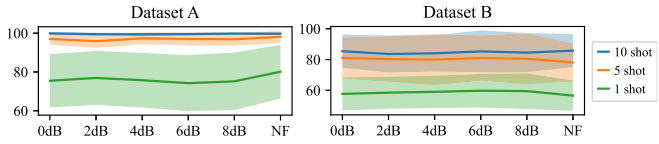


Fig. 14. Accuracy under different SNR conditions on two datasets. "NF" denotes noise-free, referring to the original signal without added noise.

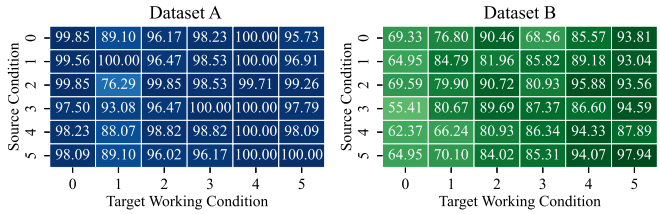


Fig. 15. Transfer learning accuracy matrices under different working conditions on two datasets.

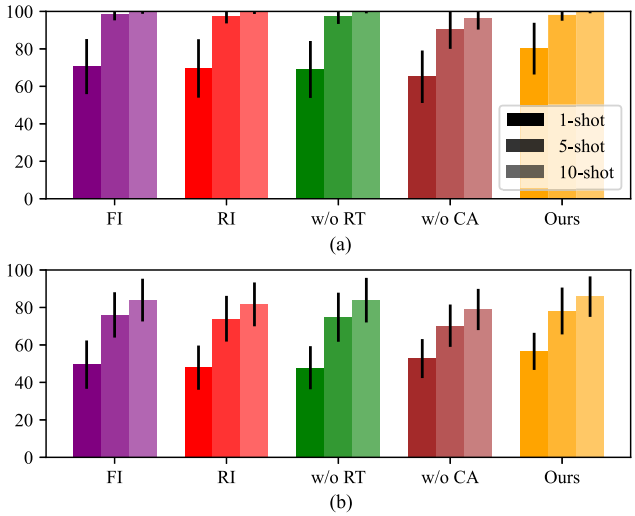


Fig. 16. Ablation study results on two datasets demonstrate the effectiveness of our proposed components. (a) Ablation study results of Dataset A. (b) Ablation study results of Dataset B

Dataset A, the accuracy under the 5-shot and 10-shot settings remains almost unaffected by noise, with only a slight accuracy drop observed under the 1-shot setting. For Dataset B, with mild noise (SNR = 8 dB), the model exhibits even higher accuracy under the 1-shot and 5-shot settings compared to the noise-free setting. This improvement may be attributed to the introduction of slight noise, which compels the model to learn more fault-related features, thereby enhancing its performance on the test set. Overall, the proposed method demonstrates strong robustness in noisy environments. Even at 0 dB, where the noise energy equals the signal energy, the model maintains diagnostic accuracy comparable to that of the noise-free setting.

We also conducted transfer learning experiments across different working conditions on two datasets to validate the generalization capability of the model using accuracy matrices. The model underwent 10-shot pretraining on the source working condition, followed by 1-shot fine-tuning on the target working condition. As shown in Fig. 15, the pretrained weights significantly improved the 1-shot accuracy under various working conditions, achieving diagnostic accuracy in some cases comparable to that of 5-shot or even 10-shot settings. This result demonstrates that the proposed method exhibits strong generalization capability, as the fault features learned from the source working condition remain highly relevant in the target working condition.

5) Ablation Study: To further validate the proposed few-shot learning enhancement methods and the CA algorithm, we conducted ablation experiments on the following two datasets.

1) *Sampling with fixed interval (FI)*, specifically, the sampling interval in the experiment is set to 2000, the same as the sample length.
2) *Sampling with random interval (RI)*, where random sampling is applied to the dataset.
3) *Without using the proposed w/o RTs*, employing only cross-entropy between predicted labels and true labels as the loss.
4) Without using the proposed w/o CA, employing equal weights of all feature channels.

The results of the experiments are presented in Tables VII, VIII, and Fig. 16. In comparison with TFSDFN utilizing optimized sampling intervals and the RT, the following conclusions can be drawn:

The optimization of sampling intervals is crucial for accuracy under few-shot settings: Under the 1-shot setting, TFSDFN achieves a 10% increase in diagnostic accuracy compared to cases with fixed or random sampling intervals. Using appropriate

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TABLE VII
ABLATION STUDY ON DATASET A

Setting	Method	C0	C1	C2	C3	C4	C5	Avg.
1-Shot	FI	73.89±10.7	49.24±4.3	76.56±7.0	67.05±9.5	90.10±4.9	66.51±9.4	70.56±14.7
	RI	74.89±8.1	48.07±7.5	73.27±7.5	65.89±11.1	91.70±5.4	63.54±9.4	69.56±15.6
	w/o RT	74.01±7.5	44.69±7.6	73.70±5.3	67.91±5.9	90.17±5.1	63.56±7.8	69.01±15.2
	w/o CA	65.15±11.9	43.98±6.1	70.85±7.2	65.71±7.8	84.65±9.4	60.44±7.4	65.13±14.0
	Ours	86.86±5.6	57.94±2.8	93.02±3.4	75.47±3.7	96.35±1.3	71.14±2.6	80.13±13.8
5-Shot	FI	99.80±0.2	91.86±2.7	99.22±0.6	99.38±0.5	100.00±0.0	99.75±0.3	98.33±3.1
	RI	99.02±0.8	90.00±1.9	97.13±1.8	98.22±0.8	99.96±0.1	98.88±1.4	97.20±3.6
	w/o RT	99.14±0.8	89.42±4.5	98.29±1.6	98.77±1.6	99.99±0.0	99.47±0.7	97.51±4.2
	w/o CA	96.48±2.0	69.26±5.0	96.39±1.9	91.48±3.5	98.21±1.7	89.50±3.6	90.22±10.2
	Ours	99.80±0.2	91.68±2.0	98.95±0.6	99.45±0.3	100.00±0.0	99.11±0.5	98.13±3.1
10-Shot	FI	99.92±0.1	98.26±1.4	99.81±0.2	100.00±0.0	100.00±0.0	100.00±0.0	99.61±0.9
	RI	99.93±0.1	97.87±0.9	99.54±0.3	99.86±0.3	100.00±0.0	99.95±0.1	99.53±0.9
	w/o RT	99.89±0.2	98.62±1.0	99.59±0.5	99.95±0.1	100.00±0.0	99.95±0.1	99.67±0.7
	w/o CA	99.11±0.4	84.62±6.1	99.01±1.0	97.78±1.9	99.26±0.6	96.89±1.3	96.11±5.8
	Ours	100.00±0.0	98.33±0.9	99.94±0.1	100.00±0.0	100.00±0.0	100.00±0.0	99.71±0.7

TABLE VIII
ABLATION STUDY ON DATASET B

Setting	Method	C0	C1	C2	C3	C4	C5	Avg.
1-Shot	FI	37.29±8.8	39.44±7.9	49.10±7.4	48.36±6.4	53.83±9.0	68.89±6.1	49.48±12.9
	RI	34.78±4.4	39.05±6.5	47.80±7.1	44.47±5.3	53.80±3.3	67.32±4.1	47.87±11.8
	w/o RT	35.11±5.1	38.20±6.2	48.72±8.2	47.22±5.4	56.03±4.6	62.70±8.7	47.84±11.5
	w/o CA	41.18±4.0	44.38±4.2	52.65±7.2	50.18±3.7	60.98±5.7	66.98±7.4	52.72±10.4
	Ours	41.40±3.5	51.00±2.5	54.85±1.8	56.00±2.9	65.55±1.5	70.55±3.6	56.56±9.9
5-Shot	FI	57.12±5.2	66.85±6.4	82.79±5.5	76.51±4.8	85.29±5.0	87.60±4.3	76.03±12.1
	RI	57.61±3.0	61.09±7.5	79.37±5.2	73.72±4.5	83.45±2.8	88.72±3.1	73.99±12.2
	w/o RT	55.50±5.7	62.56±6.1	80.97±3.3	74.69±5.7	86.67±3.6	88.34±4.8	74.79±13.1
	w/o CA	54.59±4.9	62.45±3.9	70.43±5.3	67.30±4.3	82.60±4.0	84.26±3.2	70.27±11.3
	Ours	58.52±4.1	67.27±2.4	85.13±2.2	75.61±1.8	91.79±1.4	90.43±2.2	78.12±12.5
10-Shot	FI	64.33±3.7	74.18±2.9	91.94±2.5	86.13±3.6	93.98±1.9	93.11±1.9	83.94±11.4
	RI	61.37±2.4	71.95±2.5	88.50±2.1	83.58±2.7	91.47±2.1	93.05±1.8	81.65±11.7
	w/o RT	63.74±5.1	75.47±3.5	90.64±4.5	86.59±5.5	92.87±4.2	94.11±4.8	83.90±11.9
	w/o CA	64.47±3.4	68.87±3.7	81.87±4.5	75.48±3.7	92.53±3.1	90.32±3.8	78.92±11.0
	Ours	67.13±1.9	77.95±2.4	92.76±1.8	84.53±2.3	97.63±0.8	94.66±2.0	85.78±10.8

sampling intervals, specifically the signal period, helps reduce the generation of irrelevant sample features, diminishes the difference between training and testing set samples, and enhances the diagnostic effectiveness of the model. The proposed algorithm efficiently obtains suitable sampling intervals through simple tensor operations, facilitating practical implementation in engineering.

The regularization loss plays a critical role in accuracy improvement: TFSDFN consistently outperforms the diagnosis scenarios where the regularization loss is not used in all situations. We utilized the t-distributed Stochastic neighbor embedding (t-SNE) algorithm to visualize the feature output results with and without the use of an RT during training. As shown in Fig. 17, the absence of an RT leads to a closer distribution of training sample features, which may result in potential misclassification risks on the test set. The introduction of the regularization loss enforces the model to better separate samples from different classes in the feature space during training. This approach effectively enhances the diagnostic robustness of the model, even when the training samples do not adequately represent the overall classification features, thereby further improving accuracy.

The two proposed few-shot enhancement methods have broad applicability: A comparative analysis reveals that although these two methods no longer provide significant accuracy improvements as the training samples increase, they do not lead to a decrease in accuracy. Moreover, these two algorithms do not increase the computational burden of the model, making them widely applicable to models of various sample sizes. In addition, these enhancement algorithms are structure-agnostic, making them readily applicable to any fault diagnosis algorithm.

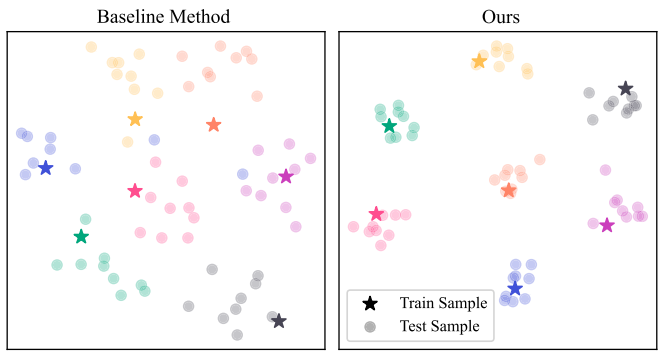


Fig. 17. Output features of different models are visualized using the t-SNE algorithm.

The introduction of the CA mechanism enhances the feature learning capability under few-shot settings: Among all ablation configurations, the removal of the CA mechanism causes the most significant drop in accuracy, particularly under extreme

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few-shot settings. This indicates that TFSDFN relies on the CA mechanism to assign weights to different sensors and convolutional scales, thereby enabling the learning of critical fault features under few-shot settings.

6) Analysis: Through the analysis of the results from the aforementioned experiments, we can draw the following conclusions.

To enhance diagnostic accuracy, especially under few-shot settings, it is imperative to utilize all available information embedded in the signals. In the aforementioned experiments, SAEDBN exhibited the poorest performance, attributed to its reliance solely on statistical features of vibration signals, leading to a substantial loss of valuable information. In contrast, ICNN, despite its simple structure, demonstrated superior performance in models utilizing time-domain signals as input, owing to the incorporation of LeakyReLU and random pooling, which mitigated the loss of crucial information. MSAWS, employing time–frequency spectrograms as input, maximized the utilization of effective information in both the time and frequency domains, surpassing models utilizing time-domain signals and statistical metrics as input. Furthermore, our proposed TFSDFN, utilizing a triple-stream network structure that integrates time-domain, frequency-domain, and statistical features, endeavors to comprehensively exploit valuable information within vibration signals. Consequently, it outperforms all comparative models under few-shot settings.

The proposed time–frequency decoupling analysis technique demonstrates a high level of superiority: While traditional time–frequency spectrogram analysis achieves effective fault diagnosis, its computational complexity is notable due to the necessity of converting time-domain signals into spectrograms as inputs. This conversion leads to all convolutional layers in the network being transformed into 2-D convolutions, substantially increasing the network's parameter count and computational load. Our proposed approach inputs both the raw time-domain signals and FFT-transformed frequency-domain signals into a dual-stream network, followed by integration using an enhanced CA mechanism. This method features a parameter and computational load significantly lower than MSAWS, providing potential for real-time diagnostics in scenarios with computational constraints. Furthermore, in terms of accuracy and robustness, the decoupled time–frequency analysis surpasses traditional spectrogram-based analysis. This is attributed to the fact that the proposed method utilizes the original time-domain signals as input, avoiding the loss of valuable information during the conversion process. The enhanced CA mechanism enables the network to more effectively extract features from the time-domain and frequency-domain signals, further enhancing the diagnostic performance of the model.

When dealing with multiple sensor inputs, assigning different weights to each sensor's input is crucial: In experiments conducted on two datasets, TFSDFN, MSAWS, and DSCNN emerged as the top three models in terms of accuracy. A common characteristic among them is the allocation of weights to multiple sensors. MSAWS and TFSDFN achieve this through attention mechanisms, while DSCNN analyzes diagnostic results for each sensor, determining confidence levels for individual sensors. This underscores the importance of assigning distinct weights to inputs from each sensor, enabling the model to effectively utilize information from each sensor. The proposed enhanced CA mechanism in this study proves to be the most effective among these methods.

V. Conclusion

With the rapid development in the industrial sectors, multisensor systems are becoming increasingly prevalent in various application scenarios. These systems, by integrating data from multiple sensors, can provide more comprehensive and accurate information, thereby enhancing the performance and reliability of systems. However, multisensor systems face numerous challenges in fault diagnosis. Addressing this issue, this article proposes a new approach for fault diagnosis with limited training samples, using a lightweight network architecture. The main innovations are as follows.

1) A triple-stream network design that separately processes time-domain, frequency-domain, and statistical features from vibration signals. This avoids complex joint time–frequency analysis.
2) Two model-agnostic techniques to enhance few-shot learning that minimize unimportant differences between samples and maximize separation between classes.

Experiments are conducted on two multisensor bearing fault datasets. Results showed the proposed method achieved higher accuracy than other methods, especially with very few (1–5) samples per fault. The lightweight design also had faster computation than approaches using time–frequency maps.

The two enhancement techniques proposed in this study can be readily applied to enhance the few-shot learning performance of any fault diagnosis algorithm. Moreover, the conceptual shift from employing a CA mechanism to fuse time- and frequency-domain information in transforming time–frequency domain 2-D convolution is also noteworthy and provides valuable insights for adaptation. In conclusion, this article presents a new fault diagnosis approach tailored for few-shot settings. By extracting complementary features and enhancing feature discrimination, the proposed approach achieves superior accuracy with minimal training data compared to existing methods. Moreover, the efficient architecture enables deployment on low-resource systems. Further testing on diverse industrial datasets would further strengthen these findings.

This study also has some limitations that will be addressed in future work. Although our method achieves high accuracy with only a small number of samples, the incorporation of transfer learning has also proven to be effective. However, our method requires predefined input channels and fault categories when constructing the model, which poses challenges for cross-dataset transfer and generalization. For instance, transferring from a 6-channel, 7-class dataset (Dataset A) to a 5-channel, 4-class dataset (Dataset B) can be problematic. In the future, we plan to conduct further research on input channel alignment and reconstruction, as well as transfer learning within a shared feature space.

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